

Requirement Analysis & Dataset Source Report

AI-Powered Real Estate Intelligence, Valuation & Investment Recommendation
Platform

Phases 1 & 2

1. PROBLEM STATEMENT

1.1 Background of the Real Estate Industry in India

The Indian real estate market is one of the most dynamic and rapidly growing sectors, generating massive volumes of data daily across various portals, government registries, and property listings. Despite this data abundance, the market remains highly fragmented and opaque. Property valuations are often driven by localized, subjective judgments rather than data-driven insights.

1.2 Challenges Faced by Investors and Buyers

Navigating the real estate market poses significant challenges for stakeholders:

- **Information Asymmetry:** Buyers and investors struggle to determine the true fair market value of a property, leading to risks of overpaying.
- **Risk Assessment:** It is difficult to objectively quantify investment risks based on localized infrastructure, market volatility, and demographic indicators.
- **Future Planning:** Estimating future appreciation, return on investment (ROI), and optimal holding periods is largely speculative.
- **Decision Paralysis:** The sheer volume of unstructured variables—ranging from amenities to location-specific metrics—makes manual comparison and decision-making overwhelming.

1.3 The Role of Artificial Intelligence

Artificial Intelligence (AI) and Machine Learning (ML) can transform raw, fragmented real estate data into actionable business intelligence. By leveraging advanced regression algorithms and explainable AI techniques, the platform can uncover hidden patterns, model complex non-linear relationships between property features and pricing, and provide mathematically grounded forecasts and recommendations.

1.4 Project Objectives

The objective of this project is to build an end-to-end AI-powered Real Estate Intelligence, Valuation & Investment Recommendation Platform. Specifically, the platform aims to:

- Predict accurate property prices based on comprehensive feature sets.
- Automatically flag properties as overvalued or undervalued relative to the market baseline.
- Calculate an objective property risk score and estimate ROI.
- Generate actionable Buy, Hold, or Sell recommendations.
- Demystify AI decisions using explainability frameworks (SHAP) to build trust with end-users.

1.5 Expected Outcomes

The final deliverable will be a fully functional, interactive Streamlit application. Users will be able to input property parameters and instantly receive a predicted price, a comprehensive risk profile, a forecasted 1-to-5-year valuation, and an explicit explanation of why the AI arrived at its conclusions.

2. SCOPE DOCUMENT

2.1 In-Scope

The project encompasses the following components, phases, and functionalities:

- **Project Phases:** Execution of all 11 defined project phases (Requirement Analysis, Data Collection, Data Cleaning, EDA, Feature Engineering, ML Modeling, Explainable AI, Recommendation Engine, Risk Score Engine, Forecasting, and Dashboarding).
- **Machine Learning:** Training, evaluation, and comparison of six specific algorithms: Linear Regression, Random Forest, XGBoost, CatBoost, LightGBM, and Gradient Boosting.
- **Core Modules:**
 - Property Price Prediction Engine.
 - Investment Recommendation Engine (Buy/Hold/Sell logic).
 - Risk Score Engine (Low/Medium/High categorization).
 - Future Price Forecasting Module (1-year, 3-year, and 5-year projections).
 - Explainable AI Module (Global and Local SHAP interpretations).
- **User Interface:** A comprehensive 9-page interactive Streamlit dashboard featuring EDA, prediction forms, investment recommendations, risk analysis, forecasting, SHAP explainability, property comparisons, and market insights.

2.2 Out-of-Scope

The following items are explicitly excluded from the current project boundaries:

- Integration with real-time API data feeds (live market updates).
- Production-grade deployment on cloud servers (e.g., AWS/GCP architecture).
- Analysis of non-Indian property markets.
- Prediction of rental yields or tenant acquisition metrics.
- Legal, regulatory, or compliance checks regarding property ownership.

2.3 Target Users

- **Real Estate Investors:** Seeking to maximize ROI and minimize capital risk.
- **Property Buyers:** Looking for fair market valuations and neighborhood insights.
- **Real Estate Agents:** Requiring data-backed justifications for property pricing.

2.4 Assumptions

- The primary Kaggle dataset is largely representative of the broader Indian real estate market dynamics.
- Synthetic and simulated data (e.g., distance to malls, crime index) generated programmatically reflect realistic market distributions for the purpose of algorithm training.
- Census 2011 population density values serve as an adequate proxy for current demographic pressure and urban density.

2.5 Constraints

- **Academic Nature:** The project is constrained by academic requirements and evaluation criteria.
- **Timeline:** Strict adherence to project deadlines.
- **Budget:** Zero-budget constraint requiring the use of exclusively free, open-source libraries, open datasets, and no paid APIs.

3. REQUIREMENT ANALYSIS REPORT

3.1 Functional Requirements

- **FR-01 (Price Prediction):** The system shall predict the precise monetary value of a property based on user-provided inputs.

- **FR-02 (Valuation Detection):** The system shall categorize a property as "Overvalued" or "Undervalued" by comparing the listed/expected price against the model's predicted baseline.
- **FR-03 (ROI Estimation):** The system shall estimate potential ROI based on historical trends and infrastructural growth scores.
- **FR-04 (Risk Scoring):** The system shall output a discrete risk category (Low, Medium, High) derived from crime indices, property age, and market variables.
- **FR-05 (Recommendation Generation):** The system shall output a deterministic "Buy," "Hold," or "Sell" signal based on a combination of ROI and Risk rules.
- **FR-06 (Explainability):** The system shall provide SHAP-based global feature importance charts and local waterfall plots explaining specific property valuations.
- **FR-07 (Forecasting):** The system shall output future price projections for 1-year, 3-year, and 5-year horizons.
- **FR-08 (Dashboard Interactivity):** The system shall feature a 9-page user-navigable Streamlit interface enabling parameter inputs and side-by-side property comparisons.

3.2 Non-Functional Requirements

- **NFR-01 (Data Volume):** The system must handle and process a minimum of 15,000 records. (The utilized dataset significantly exceeds this with 2,50,000 records).
- **NFR-02 (Model Performance):** Machine learning models must be strictly evaluated and ranked against standard regression metrics: MAE, RMSE, MSE, and R^2 Score.
- **NFR-03 (Usability):** The Streamlit dashboard must respond to user inputs within acceptable latency bounds (under 5 seconds for single property prediction).
- **NFR-04 (Transparency):** Model explainability (SHAP) must be human-readable and visually integrated into the dashboard.

3.3 Technology Requirements

- **Core Language:** Python 3.x
- **Data Manipulation:** Pandas, NumPy
- **Machine Learning:** Scikit-Learn
- **Advanced Ensembles:** XGBoost, CatBoost, LightGBM
- **Explainable AI:** SHAP
- **Visualization:** Plotly
- **Web Application Framework:** Streamlit

3.4 Data Requirements

The system requires a composite dataset built from primary listings and augmented secondary features. Required fields span Property Information (Type, BHK, Bathrooms, Floors), Location Information (City, Area), Characteristics (Area Sq.Ft, Age, Parking), Nearby Facilities (Distances to Hospitals, Schools, Metros, Airports), and Economic Indicators (Population Density, Crime Index, Infra Growth Score).

4. DATASET SOURCE REPORT

4.1 Primary Dataset Source

- **Dataset Name:** India House Price Prediction
- **Source / Platform:** Kaggle (Uploaded by user: ankushpanday1)
- **URL:** <https://www.kaggle.com/datasets/ankushpanday1/india-house-price-prediction>
- **Size:** 2,50,000 records
- **Reason for Selection:** Selected due to its massive scale (far exceeding the 15,000-record requirement), broad geographic spread (20 states, 42 cities), and rich feature set containing vital physical and location-based real estate attributes.
- **Collection Method:** Downloaded as a static CSV via the Kaggle platform.
- **Licensing:** Open-source usage as per Kaggle's community data-sharing guidelines.

4.2 Secondary Dataset Source

- **Dataset Name:** Census of India 2011
- **Source:** Government of India Official Census Portal (censusindia.gov.in)
- **Reason for Selection:** Required to fulfill the mandatory "Economic Indicators" feature (Population Density), which is highly correlated with real estate demand and pricing.

4.3 Data Integration Methodology

To merge the primary and secondary sources:

1. **Mapping:** A programmatic Python dictionary was constructed mapping City and State names from the primary Kaggle dataset to their respective population density figures derived from the Census 2011 data.
2. **Fallback Logic:** Where city-level census data was unavailable or naming conventions mismatched, a state-level median density was used as a fallback mechanism to ensure zero missing values.

3. **Synthesis:** Additional required distances and indices (e.g., Mall Distance, Crime Index) were synthetically engineered using NumPy (seed=42) constrained by city-tier logic to ensure statistically realistic data distributions.

5. DATA DICTIONARY

5.1 Comprehensive Feature Data Dictionary

Column Name	Source	Data Type	Description	Sample Value	Notes
ID	Primary	Integer	Unique identifier for each property listing.	10452	Used for indexing only.
State	Primary	String	Indian state where the property is located.	Maharashtra	
City	Primary	String	City where the property is located.	Mumbai	Key for Census merge.
Locality	Primary	String	Specific neighborhood or micro-market.	Andheri West	
Property_Type	Primary	String	Category of the property.	Apartment	e.g., Apartment, Villa.
BHK	Primary	Integer	Number of Bedrooms, Hall, and Kitchens.	3	

Column Name	Source	Data Type	Description	Sample Value	Notes
Size_in_SqFt	Primary	Float	Total built-up area of the property.	1450.0	
Price_in_Lakhs	Primary	Float	Listed price of the property in INR Lakhs.	185.50	Target Variable.
Price_per_SqFt	Primary	Float	Price calculated per square foot.	12793.10	
Year_Built	Primary	Integer	The year the property construction was completed.	2015	
Furnished_Status	Primary	String	Furnishing state.	Semi-Furnished	Unfurnished/ Semi/Fully.
Floor_No	Primary	Integer	Floor on which the unit is located.	4	
Total_Floors	Primary	Integer	Total number of floors in the building.	12	
Age_of_Property	Primary	Integer	Age of the property in years.	8	
Nearby_Schools	Primary	Integer	Number of schools within a defined radius.	3	
Nearby_Hospitals	Primary	Integer	Number of hospitals within a defined radius.	2	

Column Name	Source	Data Type	Description	Sample Value	Notes
Public_Transport_Accessibility	Primary	String	Level of access to public transit.	High	High/Medium/Low.
Parking_Space	Primary	String	Availability of parking.	Yes	Yes/No.
Security	Primary	String	Presence of 24/7 security.	Yes	Yes/No.
Amenities	Primary	String	Comma-separated list of building facilities.	Gym, Pool, Lift	Parsed later for counts.
Facing	Primary	String	Direction the main entrance faces.	East	Vaastu indicator.
Owner_Type	Primary	String	Type of seller.	Builder	Builder/Agent/Individual.
Availability_Status	Primary	String	Current market status.	Ready to Move	Under Construction/Ready.
Bathrooms	Engineered	Integer	Number of bathrooms.	2	Derived from BHK feature.
Balcony	Engineered	Integer	Number of balconies.	1	Derived via BHK-based probability mapping.
Area	Engineered	String	Standardized geographic area label.	Western Suburbs	Mapped from Locality column.

Column Name	Source	Data Type	Description	Sample Value	Notes
Population_Density	Engineered	Float	People per sq. km.	28508.0	Census 2011 city dict → state dict → median fallback.
Mall_Distance_km	Engineered	Float	Distance to the nearest shopping mall.	4.2	Simulated via city tier (NumPy seed=42).
Airport_Distance_km	Engineered	Float	Distance to the nearest airport.	12.5	Simulated per city tier (NumPy seed=42).
Crime_Index	Engineered	Float	Localized safety risk score (0-100).	42.5	Simulated with density-based mean (seed=42).
Infra_Growth_Score	Engineered	Float	Combined metric for area development.	7.5	Derived from Schools + Hospitals + Transport metrics.
Amenity_Count	Engineered	Integer	Total number of amenities offered.	3	Extracted by counting elements in Amenities column.
Price_Category	Engineered	String	Grouped bracket for property pricing.	Premium	Binned: Budget/Mid/Premium/Luxury.
Floor_Ratio	Engineered	Float	Ratio indicating height relative to building size.	0.33	Floor_No divided by Total_Floors.